



FURRING SYSTEM



Description:

The ceiling furring system is a type of ceiling in which rigid furring channels and studs are attached directly to the structural members of the building and is suitable for internal dry lining applications. This system provides a high-performance ceiling that can be used for new ceiling structures or to upgrade existing ceilings and give it a decorative look

Reference Standards:

ASTM A 653/ C1047 - Specification for Steel Sheet, Zinc Coated (Galvanized or Zinc Iron Alloy Coated) by the Hot Dip Process

ASTM C 645-09A - Standard Specification for Non-structural Steel Framing Members

ASTM C 754-99A - Standard Specification for Installation of Steel Framing Members to Receive Screw-Attached Gypsum Panel Products

BS EN 14195 - Metal Framing Components for Gypsum Plasterboard Systems

Definition of Terms:

Perimeter Angle - Channel that is fixed around the edge of the room providing a location for the Ceiling Furring Channel.

Furring Channel- a member installed perpendicularly to the main beams designed to receive screw attached gypsum board panel products.

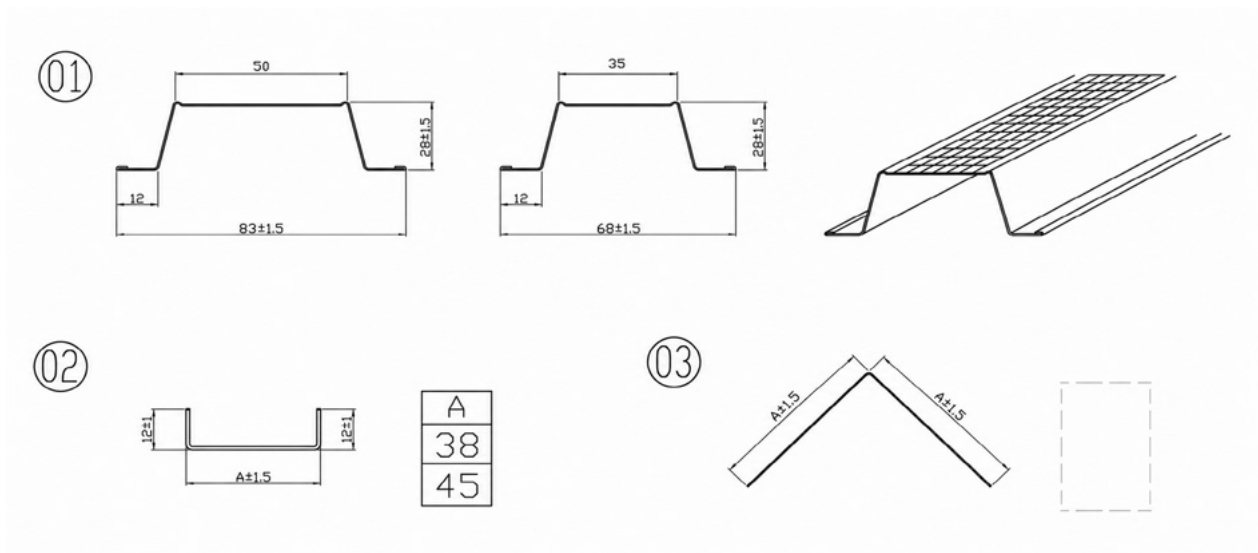
Main Channel - The channel that is attached to or suspended below the structural members of the building.

DELIVERY, HANDLING AND STORAGE:

- **Delivery:** Materials are to be delivered in original packages, containers or bundles containing the manufacturers or supplier's identification.
- **Transportation:** In transportation or lifting, handle products with care avoiding excessive stress and preventing damages.
- **Storage:** Store materials under a shaded area and keep them dry and protected against damage due to weather condition, humidity level, job site location, storage condition and surface contamination.

Material:

Profiles - pre-galvanized G60 and G90; thickness 0.4mm to 0.9mm.



Profile	Type Code	Available Thickness in mm	Length (m)	Packing pcs/ bundle
1	MAF - 35	0.45 - 0.70	3.0	500
	MAF - 50	0.45 - 0.70	3.0	500
2	MAM - 38	0.50 - 1.00	3.0	500
	MAM - 45	0.50 - 1.00	3.0	500
3	MAA - 24	0.40 - 0.50	3.0	500
	MAA - 30	0.40 - 0.50	3.0	500
	MAAW - 20	0.4 - 0.50	3.0	500

*Material: Hot Dipped Galvanized steel as per ASTM 653-Zinc Coating Z180-2750



Accessories:

Accessories	Dimensions
Gypsum Board	Thickness = 9.5mm, 12.5mm, 15mm, 18mm
Adjustable rod hanger	Rod Diameter = 3.8mm
Hanger Bracket	Thickness = 0.5 mm
M6 Threaded Rod	Length = 1m, 2m, 3m
U Bracket	Thickness = 0.5mm
Drywall Screw	Length = 25mm, 35mm, 45mm
Furring Connection Clip	Diameter 2.5 mm
Drywall Joint Tape	W = 50 mm

Installation:

1-Preparation:

1. Installation site should be kept clean and dry.
2. It is recommended that a sketch of the layout should be prepared marking all the necessary dimensions and taking into consideration the doors and windows.

3. Calculation Framing Requirements

A. Calculate the number of main channels needed for a suspended ceiling, take the shortest span of the ceiling (in meters) and divide it by 1.2 m (based on ASTM C754-99a, which limits spacing between main channels to a maximum of 1240 mm). Round up to get the number of main channels required across the ceiling. Then multiply by the length of the room in the direction of the channel to get the total linear meters of main channel required. Main Channels are normally sold in 3 meters length)

B. Calculating the Furring Channels Needed: In consideration to the linear measurement of ceiling in meters, take the longest length and divide it by 0.6m - spacing requirement - (as per ASTM C754-99a where Furring Channel = 19mm; max. span = 1240mm) to get the total number of main channels needed. Then, multiply the result by the length of the room in the direction of each furring channel to get the total linear meters of furring channel required. K5 Furring Channels are sold in 3 meters length.

C. Calculate the full perimeter of the ceiling area. This is done by adding together the lengths of all four sides of the room. K5 perimeter angles are normally sold in 3 meters length

Product Data Sheet

K5-Gypsum Board Access

REV 00, July 2026

TDS-03-01-GBA



2- Installation Method of Metal Furring System:

Installation shall meet all requirements stated on the project specification and other local statutory requirements.

1. Site Preparation:

- Clear the area and ensure the structure (walls, slab) is ready.
- Verify dimensions, check for levelness, and identify services (electrical, HVAC).

2. Marking the Ceiling Height:

- Use a laser level or water level to mark the desired ceiling height on all walls.
- Snap a chalk line around the room to indicate where the perimeter angle will be fixed.

3. Install Perimeter Angle (Wall Angle):

- Fix perimeter angles along the marked line on all walls using appropriate fasteners.
- Make sure the perimeter angle is level all around.

4. Install Hangers (Suspension Wires/Rods):

- Fix hanger wires to the structural soffit above using anchors or hooks.
- Space them typically 1.2 meters apart in both directions, forming a grid to support the main channels.
- Use adjustable wire hangers or double nuts for leveling.

5. Install Main Channels:

- Suspend main channels (38mm wide) from the hanger wires, spaced 1.2 meters apart.
- Use a spirit level or laser level to ensure they're perfectly level.

6. Install Furring Channels (Cross Channels):

- Fix furring channels perpendicular to the main channels at 600mm spacing.
- Clip them into the main channels using appropriate fasteners or locking clips.
- Ensure alignment and flatness throughout.

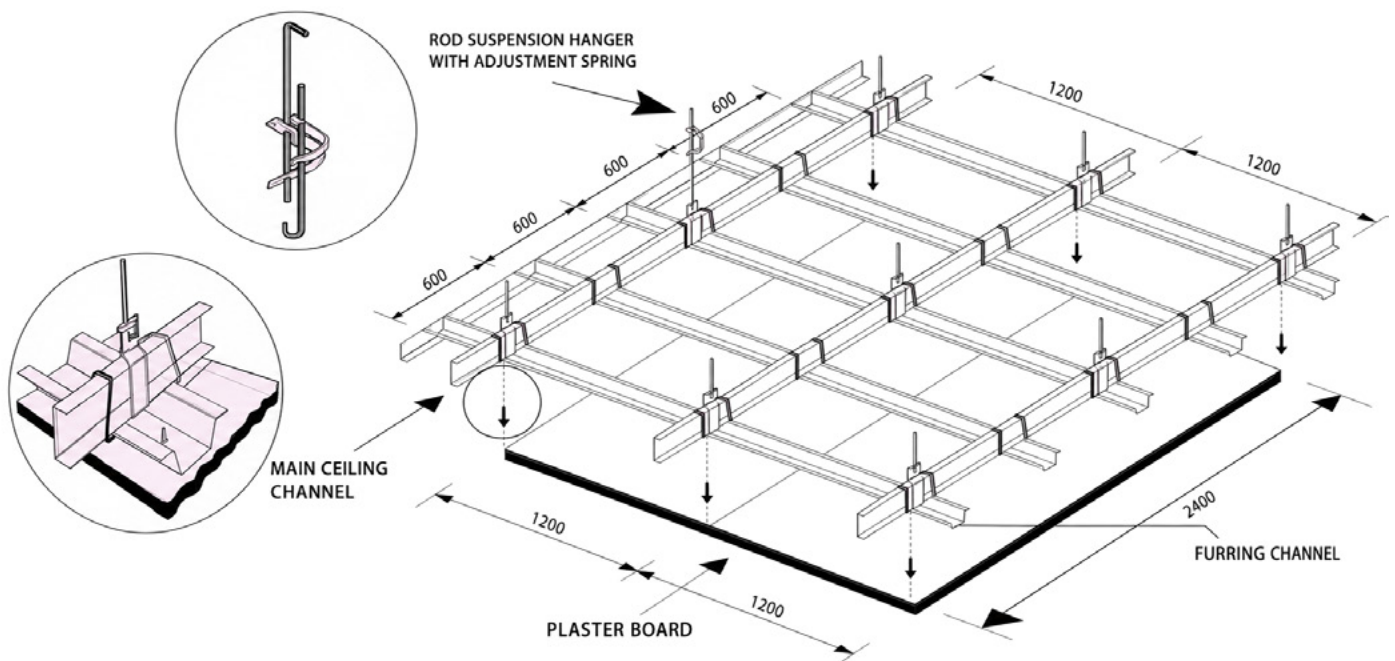
7. Install Ceiling Boards or Panels:

- Lay or screw ceiling boards onto the furring channels.
- Leave access panels if necessary for maintenance.

8. Finishing:

- Apply joint compound and tape for gypsum board systems.
- Prime, and paint if required.
- Clean the work area and inspect alignment, level, and aesthetics.

Furring Ceiling System



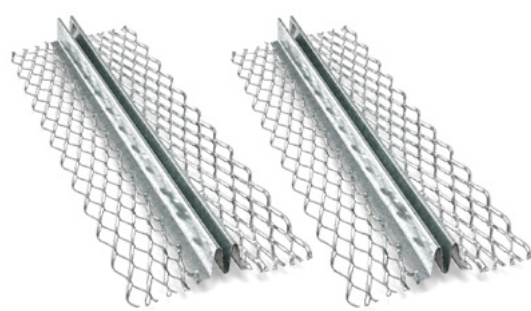


Control Joint

Description:

GI Control Joint 10×13mm aligned with a 10mm ceiling reveal. This product meets ASTM C1047 requirements and serves as an ideal baseline. The control joint bead is a continuous galvanized steel strip folded (M shape) to allow expansion and contraction.

Material: High-quality pure zinc coil steel—excellent corrosion resistance.



Expansion Joint

Description:

Aluminum Expansion Joint aligned with a 30 mm ceiling reveal. Can be used for the ceiling tiles, gypsum board with interchangeable insert & high movement absorption.

Material: Aluminum Extrusion Grade 6063 Alloy ASTM B 221 M

WIDTH (mm)	HEIGHT (mm)	TOTAL WIDTH (mm)	LENGTH (m)	H MOVEMENT	MATERIAL
30	45	70	3	10 (+/-5)	Alum + Rubber
Complies with ASTM E 1399 Cyclic movement requirement for expansion joint.					

Material: Aluminum Extrusion Grade 6063 Alloy ASTM B 221 M



A brand of

RBUILD

K5

DRY WALL PARTITION



Description:

Dry wall is an interior wall or partition finished in a dry material, usually in the form of prefabricated sheets or panels fastened to galvanized studs and tracks.

Reference Standards:

ASTM A 653/A 653 M-Specification for Steel Sheet, Zinc-Coated (Galvanized or Zinc- Iron Alloy Coated (Galvannealed) by the Hot-Dip Process.

ASTM C 645-09A - Standard Specification for Non-structural Steel Framing Members.

ASTM C 754-99A - Standard Specification for Installation of Steel Framing Members to Receive Screw-Attached Gypsum Panel Products.

BS EN 14195 - Metal Framing Components for Gypsum Plasterboard Systems.

Definition of Terms:

Framing Member - metal studs, runners (track) designed to receive screw attached gypsum panel products.

Runner a member designed to receive the ends of metal stud, attached directly to the structural members of the building.

Stud-vertical member in the frame work to which panel boards are fastened.

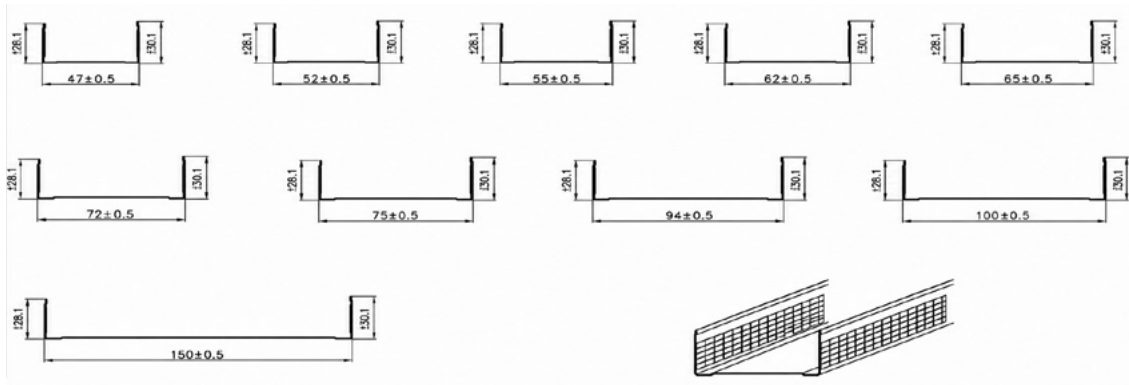
DELIVERY, HANDLING AND STORAGE:

- **Delivery:** Materials are to be delivered in original packages, containers or bundles containing the manufacturers or supplier's identification.
- **Transportation:** In transportation or lifting, handle products with care avoiding excessive stress and preventing damages.
- **Storage:** Store materials under a shaded area and keep them dry and protected against damage due to weather condition, humidity level, job site location, storage condition and surface contamination.



Material:

A. Runner pre-galvanized; thickness 0.45mm to 1.0mm



Type Code	Available Thickness in mm	Standard Length in mm	Length (m)	Packing pcs/Blt
MAT 42	0.45 - 1.00	28-30 (+/-1)	3.0	500
MAT 47	0.45 - 1.00	28-30 (+/-1)	3.0	500
MAT 52	0.45 - 1.00	28-30 (+/-1)	3.0	500
MAT 55	0.45 - 1.00	28-30 (+/-1)	3.0	500
MAT 62	0.45 - 1.00	28-30 (+/-1)	3.0	500
MAT 65	0.45 - 1.00	28-30 (+/-1)	3.0	500
MAT 72	0.45 - 1.00	28-30 (+/-1)	3.0	500
MAT 75	0.45 - 1.00	28-30 (+/-1)	3.0	500
MAT 94	0.45 - 1.00	28-30 (+/-1)	3.0	450
MAT 100	0.45 - 1.00	28-30 (+/-1)	3.0	450
MAT 150	0.45 - 1.00	28-30 (+/-1)	3.0	450

*Other lengths can be manufactured as per design.

*Material: Hot Dipped Galvanized steel as per ASTM 653-Zinc Coating Z180-2750

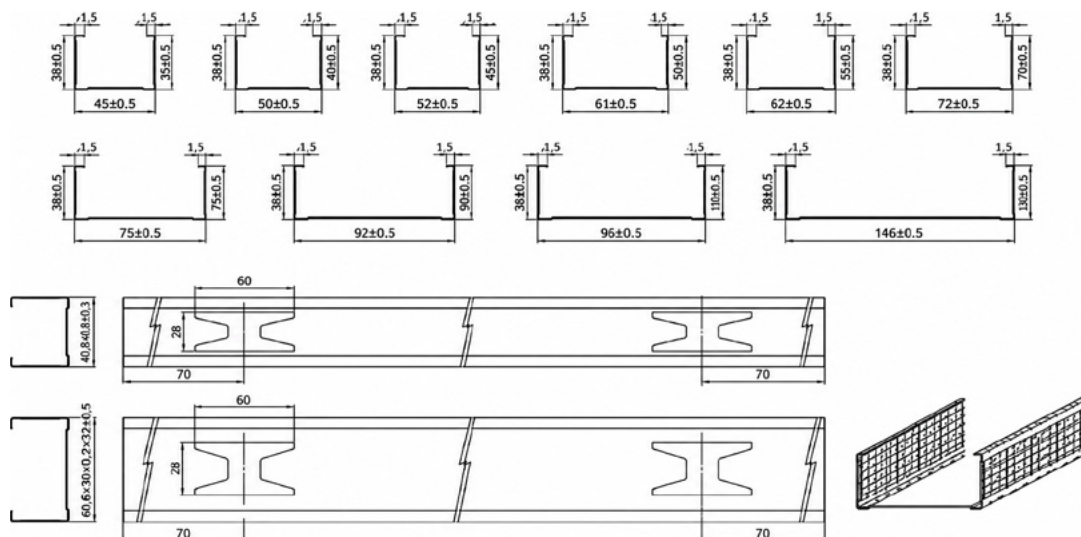
Product Data Sheet

K5-Gypsum Board Access

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B. Stud pre-galvanized G60 and G90; thickness 0.45 to 1.0mm



Type Code	Available Thickness in mm	Standard Length in mm	Length (m)	Packing pcs/Blt
MAS 40	0.45 - 1.00	34-36 (+/-0.5)	3.0	500
MAS 45	0.45 - 1.00	34-36 (+/-0.5)	3.0	500
MAS 50	0.45 - 1.00	34-36 (+/-0.5)	3.0	500
MAS 53	0.45 - 1.00	34-36 (+/-0.5)	3.0	500
MAS 60	0.45 - 1.00	34-36 (+/-0.5)	3.0	500
MAS 63	0.45 - 1.00	34-36 (+/-0.5)	3.0	500
MAS 70	0.45 - 1.00	34-36 (+/-0.5)	3.0	500
MAS 73	0.45 - 1.00	34-36 (+/-0.5)	3.0	500
MAS 92	0.45 - 1.00	34-36 (+/-0.5)	3.0	450
MAS 98	0.45 - 1.00	34-36 (+/-0.5)	3.0	450
MAS 148	0.45 - 1.00	34-36 (+/-0.5)	3.0	450

*Other lengths can be manufactured as per design.

*Material: Hot Dipped Galvanized steel as per ASTM 653-Zinc Coating Z180-2750



C. Corner Bead pre-galvanized; thickness 0.35 – 0.45mm

	Type Available Thickness Length (m)		
	Wall Angle	0.35 mm - 45 mm	3.0

D. Edging Bead pre-galvanized; thickness 0.35 – 0.45mm

	Type Available Thickness Length (m)		
	J Bead	0.35 mm - 45 mm	3.0

E. Accessories

Accessories	Dimensions
Gypsum Board	Thickness = 12.5mm & 15mm
Drywall Screw	Length = 25 mm
Self-Screw	Length = 9 mm
Drywall Joint Tape (Fiber Mesh)	50 mm

Installation

1- Preparation:

1. Installation site should be kept clean and dry.
2. It is recommended that a sketch of the layout should be prepared marking all the necessary dimensions and taking into consideration the doors and windows. Partition layouts should be marked accurately at the site using plumb bob, marker and chalk line
3. Calculating Stud and Track Requirements
 - A. Calculate Studs: Based on spacing requirements (ASTM C754-99A), divide the wall length by 600mm (where standard = max. 610mm) in order to calculate the number of studs needed. Add studs for corners and openings.
 - B. Calculating Track: Take the total linear measurement of the wall in meters and multiply by 2 to get the total requirement for floor and ceiling. Track is sold in 3 meter lengths (other lengths can be supplied upon request and consultation with our production department. Add track for door and window headers and sill.

2- Installation of framing system:

Installation shall meet all requirements stated on the project specification and other local statutory requirements.

- A. Cut studs and track to required lengths.
- B. Position tracks starting at the bottom, then to the wall going to the top. Attach tracks using appropriate fastener for the type of material you are attaching the track to (e.g. concrete, steel or timber)
- C. Insert stud at an angle and twist to lock as shown in the figure below. The friction fit will hold the stud in place. Space studs at 600mm ((where standard = max. 610mm). Make sure that all the studs are facing in the same direction. For corners, junctions and wall ends, it is recommended to use extra studs and “boxing” the studs as shown in Figure 2. Please note that there are several opinions raised in securing the studs to the tracks, although it is not stated in ASTM, it is in the designers and contractors discretion.

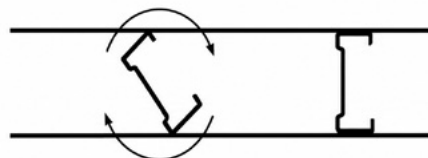


Figure 1 – Installation of stud

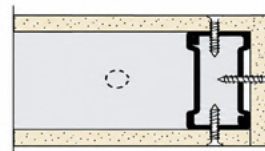


Figure 2 – Wall End (Box Stud)

D. Headers and sill - simply fabricated by cutting a piece of track 6" to 12" longer than the opening. The track is centered in the opening with the excess cut at the vertical legs, and bent 90°, this U-channel metal profile will then be fastened at the correct head or sill height. Note that there also pre-manufactured headers and sills especially used for large projects like hotels.

2-Installation of Gypsum boards:

As the face of steel stud can deflect initially during fixing, thus there should be a correct fixing technique to be followed.

The first board shall be attached to the open side of the stud. Support the stud to avoid twisting and fully screw the board to this stud before continuing to the next. The next sheet can now be screwed to the closed side of the stud. Deflection will be minimal as the first sheet will keep it rigid thus achieving a flat joint.

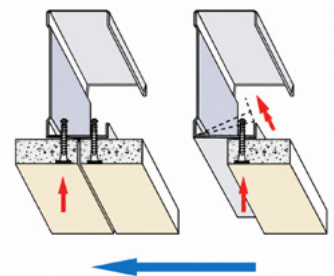
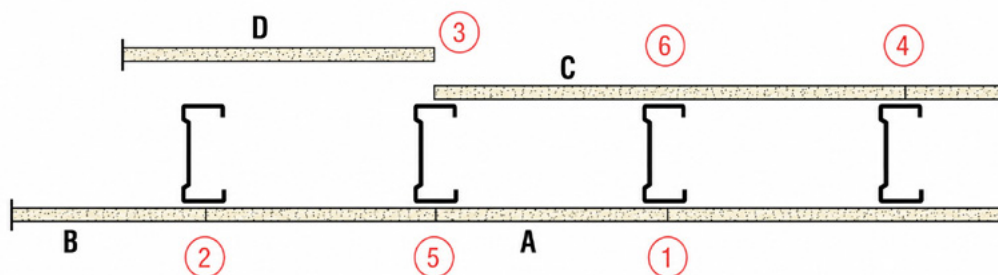


Figure 5 - Correct Fixing Direction

Board Fixing Sequence

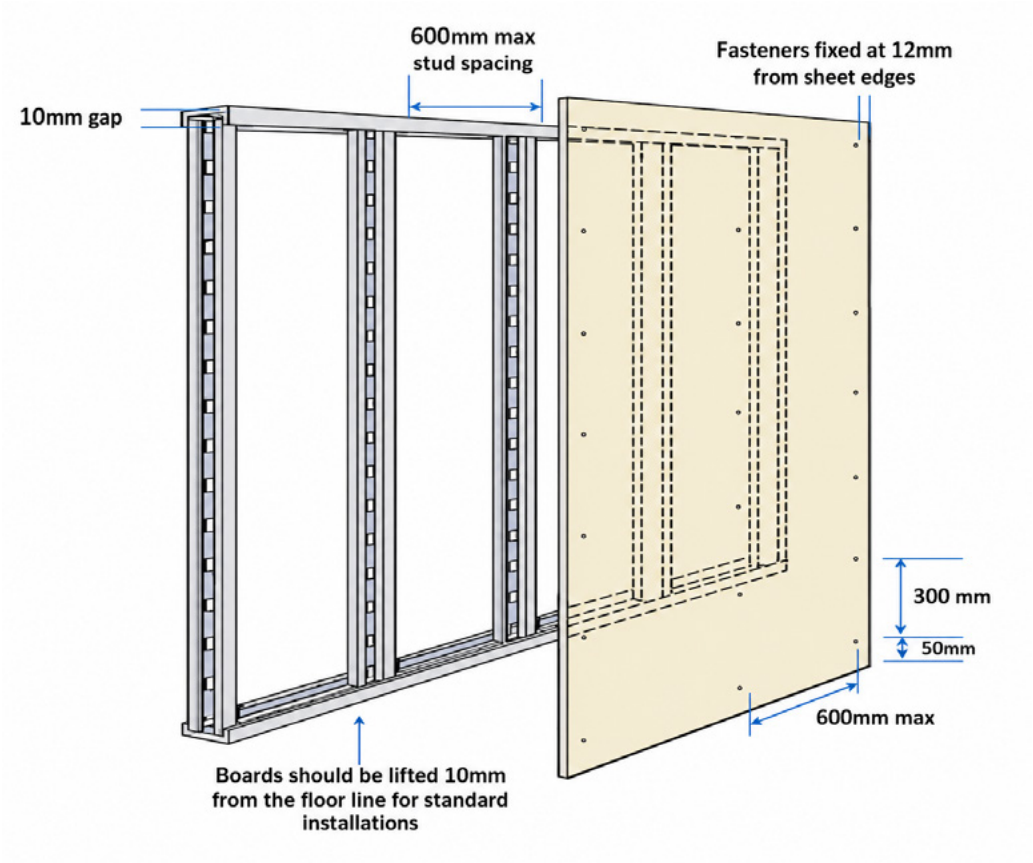
As the face of steel stud can deflect initially during fixing, thus there should be a correct fixing technique to be followed.



The boards shall be fixed in the sequence shown ABCD. When installing the first side (A&B), fasten the board at the edges only (1&2). The center must not be fastened until the second side (C&D) have been installed. Locate board joints on each side of the wall on alternate studs. Fasten the boards in sequence 1-6. This will minimize misalignment and give a higher quality finish.



Installation Details



Installation Details-Continued

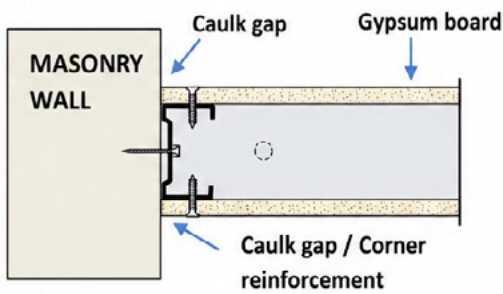


Figure 7 - INTERSECTION - Masonry

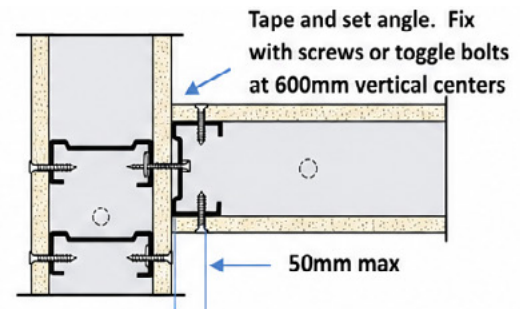


Figure 8 - INTERSECTION - Single Layer Application

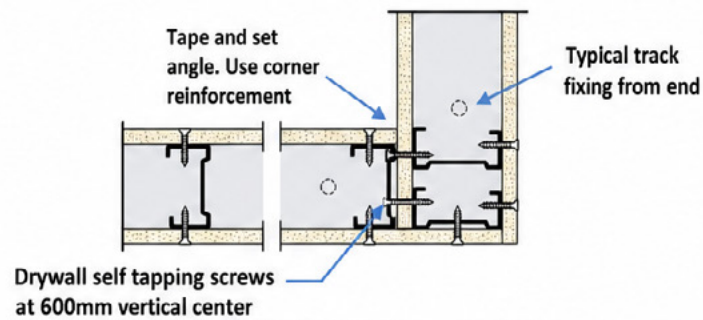


Figure 9 - CORNER - Single Layer Application

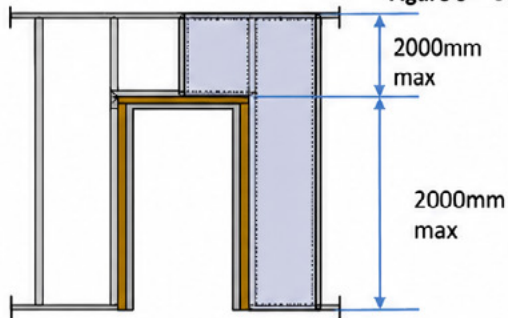


Figure 10 - Single Door

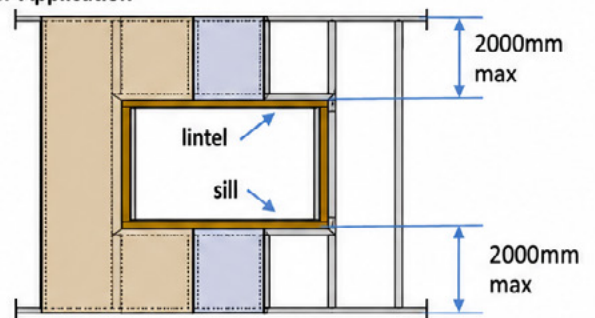


Figure 11 - Windows